

Designing Android App for Communication with Bluetooth Module

This article is about designing an Android app for communication with HC-05 Bluetooth module using MIT App Inventor (open source), a block-based programming tool. MIT App Inventor is used to create apps without any text-based programming language or bulky software in your PC.

Ardduino Uno board along with a Bluetooth module and an LED is used for testing the app.

MIT App Inventor

Open MIT App Inventor in Google Chrome or Mozilla Firefox—not supported in Internet Explorer. A

For example, here we used HC05_with_Phone (Figure 3).

The project window will open (Figure 4). Here, you will see two headings on the top-right, namely, Designer and Blocks. Under Designer head you find windows such as Palette, Viewer, Components and Properties, which are used to design an app. The user interface is designed using Designer, where you can select the various components for the app.

Blocks editor provides the functionality for the components. Main programming is done in Blocks section because MIT App Inventor is block-based programming.

App Inventor Designer section

Details of each window in Designer are given below.

Palette: Here, you can pick the different components you want to use for your project.

Viewer: Here, you can see what your app will look like.

Components: This is where you can see all the components you have used in your project.

Properties: Here, you can change the properties of each component you used, including size, colour, default text and so on.

Now, let us learn about the horizontal and vertical arrangement layouts in Android apps.

Vertical arrangement layout:

When you use vertical arrangement, all components for the user interface—such as buttons, labels and edit text—will be arranged vertically, or laid out from top to bottom.

Horizontal arrangement layout:

When you use horizontal arrangement, all components will be arranged horizontally, or left-to-right fashion.

There is one vertical and five horizontal arrangements in this project.

Vertical arrangement: Select VerticalArrangement component from the left side and drag it to Viewer

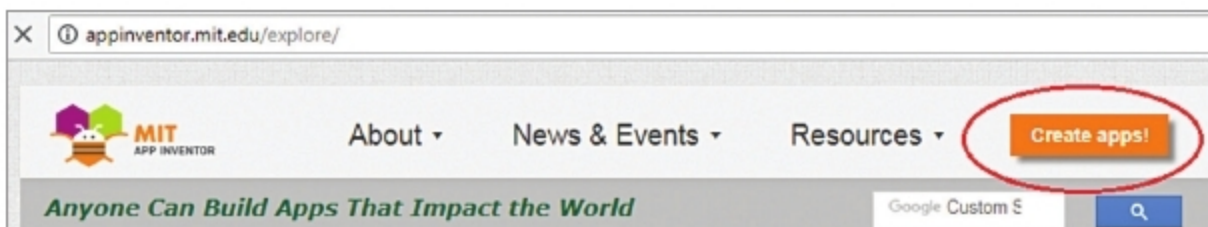


Figure 1: Create apps

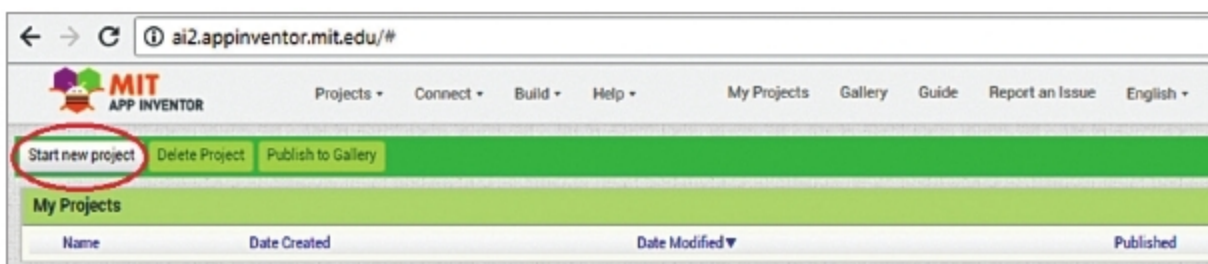


Figure 2: Start new project

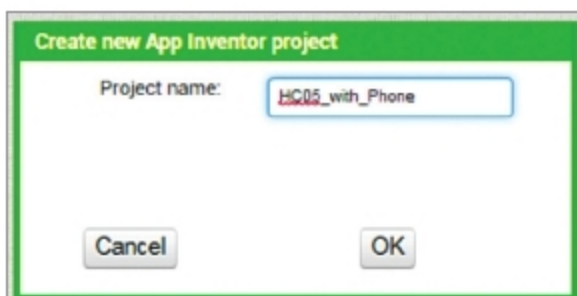


Figure 3: Name of the project

window will open, as shown in Figure 1.

Click on *Create apps!* And you will reach the login page of Google account. Sign in with your Google credentials. After signing in, a new window will open (Figure 2). Click on *Start new project*. A dialogue box will pop up for you to enter the name of your project.

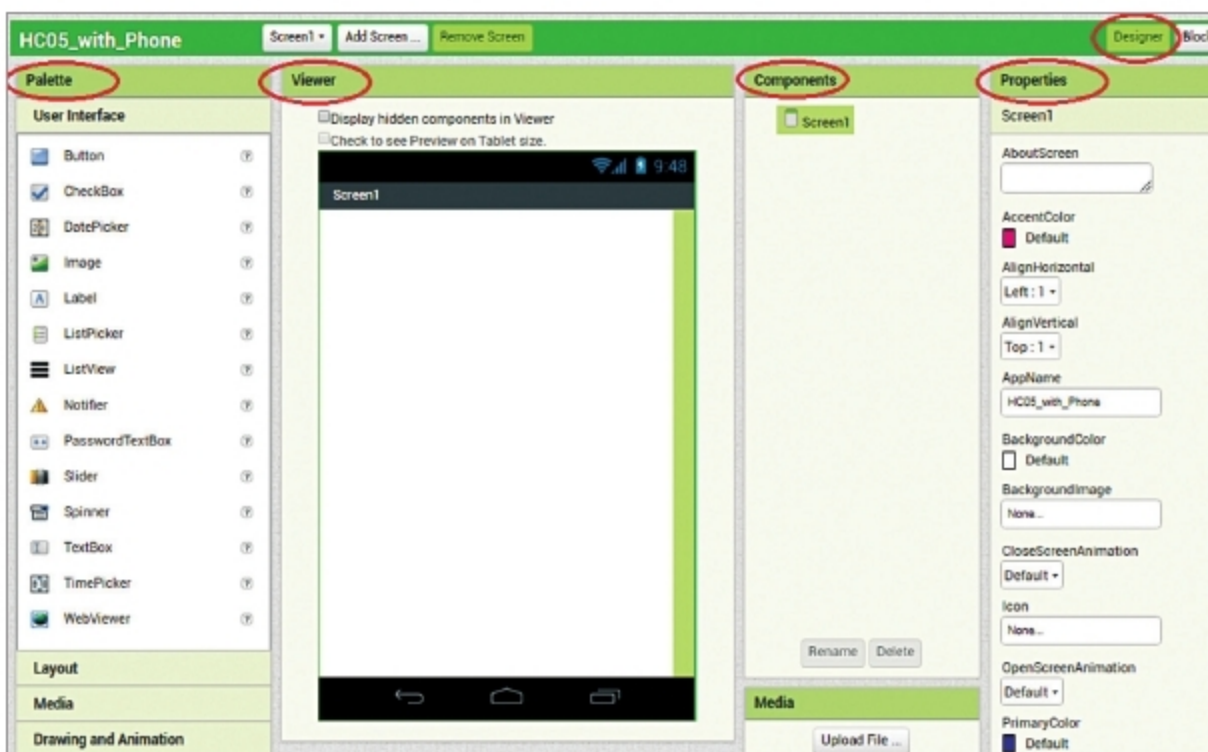


Figure 4: Designer project window